

AI Career Intelligence Platform - Project Blueprint

Overview

AI-powered platform that helps users analyze resumes, find jobs, identify skill gaps, predict salary insights, chat with an AI career assistant, and generate personalized learning roadmaps.

Phase 1 – MVP

1. Resume Upload (PDF parsing with PyMuPDF)
2. Target Role Input
3. Skill Extraction (spaCy, SkillNER)
4. Job Description Collection
5. Job Match Score using embeddings and cosine similarity
6. Skill Gap Analysis
7. Dashboard (Next.js, Tailwind, Recharts)

Phase 2 – Machine Learning Layer

Datasets:

- Resume datasets
- Job description datasets
- O*NET skills data

Models:

- Job Match Prediction (Logistic Regression, Random Forest, XGBoost)
- Salary Prediction (XGBoost Regressor, Random Forest Regressor)
- Skill Gap Prediction

Phase 3 – AI + RAG

Knowledge Base:

- Career guides
- Interview questions
- Learning resources
- Company information

Vector Database:

- ChromaDB

Pipeline:

User Query -> Embedding -> ChromaDB Search -> Retrieved Documents -> LLM -> Response

Learning Roadmap Generator:

Dynamic skill-learning plans based on target career role.

Architecture

Frontend (Next.js)

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Backend (FastAPI)

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PostgreSQL

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ChromaDB

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ML Models

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RAG Pipeline

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LLM API

Recommended Tech Stack

Frontend: Next.js 15 + Tailwind

Backend: FastAPI

Database: PostgreSQL

Vector DB: ChromaDB

Authentication: JWT

Queue: Celery + Redis

Resume Parsing: PyMuPDF

NLP: spaCy

Embeddings: Sentence Transformers

ML: Scikit-Learn + XGBoost

LLM: OpenAI or Gemini

Deployment: Docker

CI/CD: GitHub Actions

Monitoring: Prometheus + Grafana

Recruiter Value

- Data Engineering

- Data Science

- Machine Learning

- NLP

- RAG
- Vector Databases
- LLM Integration
- Backend Development
- Frontend Development
- MLOps
- Cloud Deployment